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Roll No

AU-702(A)-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Renewable Energy System

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss about beam and diffuse radiation. Discuss about solar constant.
b) What are the main components of a flat plate collector?
2. a) What is Bio-mass? How can they be converted to Bio-fuel?
b) Compare bio-mass gasification and bio-gas generation? Draw any one type of bio-gas digester.
3. a) How can wind energy be converted into electrical energy? Compare stand alone and grid connected wind energy system.
b) Describe the main applications of wind energy giving neat sketches.
4. a) What do you understand by biomass energy? Classify various biomass energy conversion technologies. 7
b) Explain the principle of generation of power using biomass fuel.

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5.
 - a) Describe with a neat sketch the Deenbandhu biogas plant. State its advantages.
 - b) Describe with a neat sketch the fluidised gasifier. State its advantages.
6.
 - a) Explain different types of turbines used in small hydro, mini hydro and micro hydro electric plants.
 - b) Explain the site selection and components of small hydro system.
7.
 - a) How is geothermal energy used for generation of electrical power? Write its advantages and disadvantages.
 - b) Explain the principle and working of fuel cell.
8. Write short notes (any two):
 - a) Hybrid systems
 - b) Wind energy storage system
 - c) Solar collectors
 - d) Biomass plant
